

Networking Fundamentals

Networking concepts, diagnostic practice & Cisco Packet Tracer implementation.

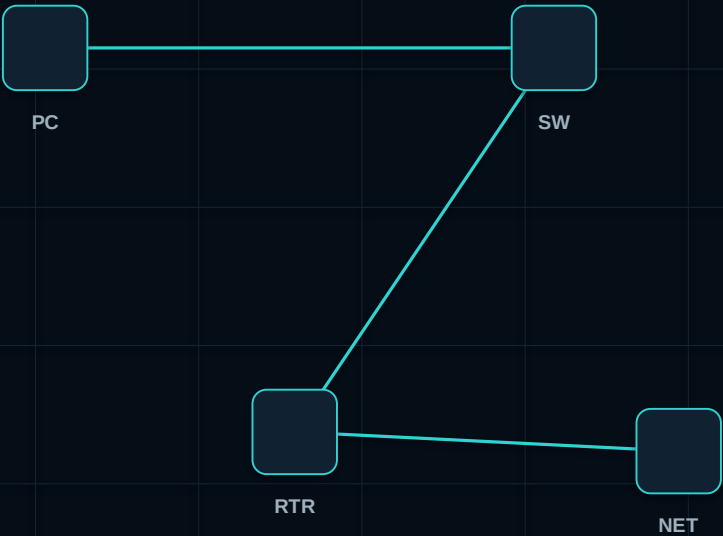
WEEK 01

STUDENT

SAMYUKTHA S

INTERN ID

DG/SEPTEMBER/CYBER/102



MODELS / PROTOCOLS / ADDRESSING / DIAGNOSTICS

01 / SCOPE

Week 1 objectives

A compact route from network architecture to diagnostics and a simulated LAN.

01 OSI MODEL

Understand the seven-layer reference model and layer roles.

02 TCP/IP MODEL

Relate the four-layer TCP/IP view to OSI.

03 PROTOCOLS

Study HTTP, HTTPS, FTP, DNS and DHCP.

04 IPv4 + CIDR

Read addressing, masks and basic subnetting.

05 DIAGNOSTICS

Use ping, ipconfig, ifconfig, tracert and traceroute.

06 PACKET TRACER

Build, configure and verify a small PC → switch → router LAN.

WHAT I LEARNED

The week connected architecture, protocols, addressing and practical diagnostics into one view of how devices communicate.

02 / ARCHITECTURE

OSI & TCP/IP models

Two complementary ways to structure network communication.

OSI / 7 LAYERS

7	APPLICATION	HTTP • HTTPS • FTP • DNS • DHCP
6	PRESENTATION	Encoding • encryption • compression
5	SESSION	Establish • maintain • terminate
4	TRANSPORT	TCP • UDP • ports
3	NETWORK	IP • packets • routers
2	DATA LINK	MAC • frames • switches
1	PHYSICAL	Bits • signals • media

TCP/IP / 4 LAYERS

APPLICATION	HTTP • HTTPS • FTP • DNS • DHCP	OSI 7–5
TRANSPORT	TCP • UDP	OSI 4
INTERNET	IP addressing • routing	OSI 3
NETWORK ACCESS	Ethernet/Wi-Fi • frames • physical transmission	OSI 2–1

WHAT I LEARNED

The models give a structured mental map: communication can be separated into layers rather than treated as one large process.

03 / PROTOCOLS

Networking protocols

Purpose, port and the key security detail recorded in the Week 1 material.

PROTOCOL	PURPOSE	PORT	KEY DETAIL
HTTP	Web communication using request/response	80	Traditional HTTP has no TLS-based cryptographic protection.
HTTPS	HTTP protected with TLS	443	TLS provides confidentiality and integrity protection and can authenticate the server.
FTP	File transfer between client and server	21	Traditional FTP lacks built-in TLS protection. FTPS uses TLS; SFTP operates over SSH.
DNS	Resolves domain names to IP addresses	53	Commonly UDP; TCP can also be used.
DHCP	Automatically provides network configuration	67/68	DORA: Discover → Offer → Request → Acknowledgment.

WHAT I LEARNED

I learned that HTTP and HTTPS serve the same broad web communication purpose, but HTTPS adds TLS protection. Traditional FTP does not provide built-in TLS protection.

04 / ADDRESSING

IPv4 addressing & subnetting

From address structure to the documented /27 example.

IPv4 FUNDAMENTALS

IPv4	32-bit address • four decimal octets
PRIVATE RANGES	10.0.0.0/8 • 172.16.0.0/12 • 192.168.0.0/16
STATIC	Manually assigned and normally predictable
DYNAMIC	Usually assigned by DHCP and may change
LOOPBACK	127.0.0.1 • local host / localhost
OBSERVED	Windows VMnet1: 192.168.45.1/24 Kali eth0: 192.168.56.129/24

CIDR / SUBNETTING

192.168.5.140 /27

/27 → 27 network bits + 5 host bits → 32 total addresses

NETWORK	HOST RANGE	BROADCAST
192.168.5.128	192.168.5.129 – 192.168.5.158	192.168.5.159

DOCUMENTED RESULT

NETWORK	USABLE HOSTS	BROADCAST
192.168.5.128	129–158 30 normally usable	192.168.5.159

WHAT I LEARNED

CIDR notation makes the network/host split explicit; the /24, /26 and /27 examples connect prefix length to address capacity.

05 / DIAGNOSTICS

Network commands

Each command answers a different practical diagnostic question.

ping

WINDOWS / LINUX

Reachability + round-trip time

Showed whether a destination responded and provided loss/timing information.

ipconfig /all

WINDOWS

Adapter + DHCP + DNS detail

Helped identify interface addressing and configuration details.

ifconfig

KALI / LINUX

Interface inspection

Helped inspect interface addresses and loopback information.

tracert

WINDOWS

Trace network hops

Showed path behavior and that some hops may not return a response.

traceroute

KALI / LINUX

Trace hops using increasing TTL

Showed how path visibility can differ when intermediate probes do not respond.

WHAT I LEARNED

ping checks reachability; ipconfig/ifconfig expose configuration; tracert/traceroute show path behavior. Running them made the theory concrete.

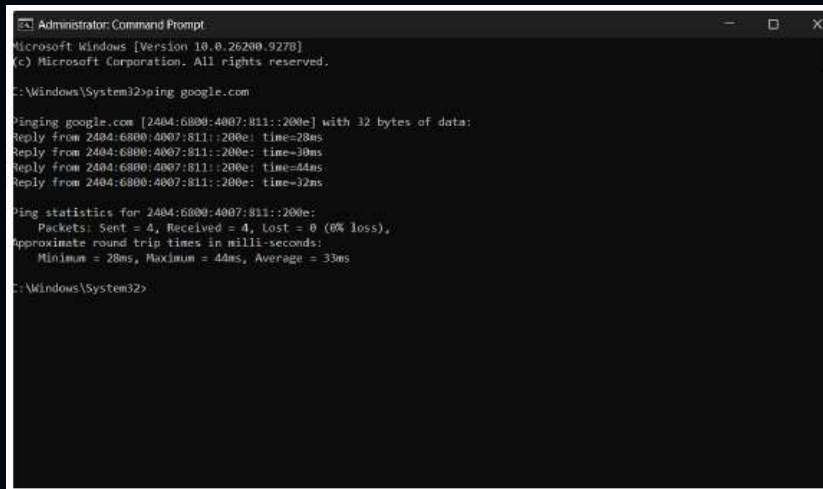
DIAGNOSTIC VIEW

REACHABILITY → CONFIGURATION → PATH

06 / EVIDENCE

Windows diagnostic evidence

Actual command outputs supplied with the Week 1 report.



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.9278]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>ping google.com

Pinging google.com [2404:6800:4007:811::200e] with 32 bytes of data:
Reply from 2404:6800:4007:811::200e: time=28ms
Reply from 2404:6800:4007:811::200e: time=30ms
Reply from 2404:6800:4007:811::200e: time=44ms
Reply from 2404:6800:4007:811::200e: time=32ms

Ping statistics for 2404:6800:4007:811::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    approximate round trip times in milli-seconds:
        Minimum = 28ms, Maximum = 44ms, Average = 33ms

C:\Windows\System32>
```

FIGURE 01 — Actual Windows ping evidence: google.com resolved and returned four replies with 0% packet loss; the observed destination was IPv6.



```
ipconfig /all

Ethernet adapter VMnet1:

   Media . . . . . : 10/100/1000 Mbps
   Link-local IPv6 . . . . . : fe80::5056:dc14:3e9d:599a%1
   IPv4 Address. . . . . : 192.168.45.1
   Subnet Mask . . . . . : 255.255.255.0
   Default Gateway . . . . . : 192.168.45.1
   .
   .
   .
   DNS Servers . . . . . : 192.168.1.1
   .
   .
   .
   DHCP Enabled. . . . . : Yes
   IP Address from DHCP . . . . . : 192.168.45.1
   Lease Obtained . . . . . : 2024-09-10 10:10:10
   Lease Expires . . . . . : 2024-09-10 10:10:10
```

FIGURE 02 — Actual Windows ipconfig /all evidence: VMware VMnet1 recorded 192.168.45.1/24, DHCP enabled, with DNS information including 192.168.1.1.

OBSERVATION

Hostname resolved;
four ICMP replies received;
packet loss recorded as 0%.

WHAT I LEARNED

ping can combine name resolution with a basic reachability test and provide reply information such as response time.

VMNET1

192.168.45.1

/24 • DHCP enabled

DNS OBSERVED

192.168.1.1

07 / EVIDENCE

Kali Linux diagnostics

Actual interface and traceroute outputs — observations only, without inferring causes.

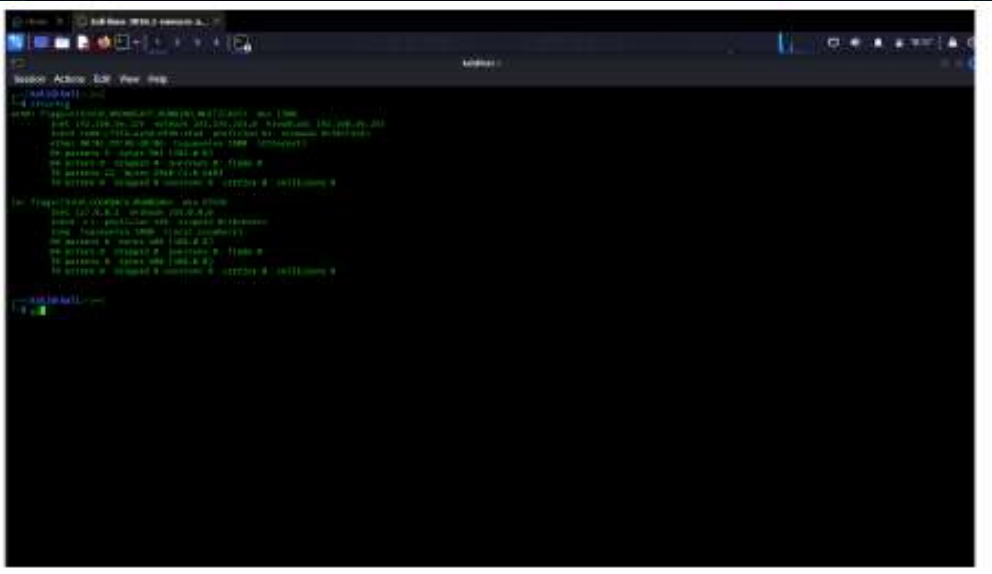
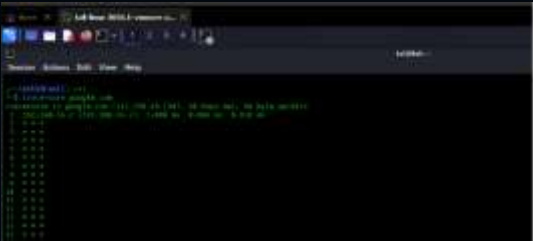


FIGURE 03 — Actual Kali Linux ifconfig evidence: eth0 recorded 192.168.56.129/24 and lo recorded 127.0.0.1.
FIGURE 05 — Actual Kali Linux traceroute evidence: hop 1 responded with 192.168.56.2; hops 2–30 showed *** and the destination was not identified within the 30-hop limit.



IFCONFIG / OBSERVATION

ETH0

192.168.56.129
/24 mask

LOOPBACK

127.0.0.1

The command output made the interface address and local loopback visible directly in the terminal.

WHAT I LEARNED

Only the first hop was visibly responsive. The asterisks are evidence of no visible response for those probes, not proof of a failed router.

08 / PRACTICAL

Cisco Packet Tracer practical

A small LAN: PC → 2960-24TT switch → 2911 router.

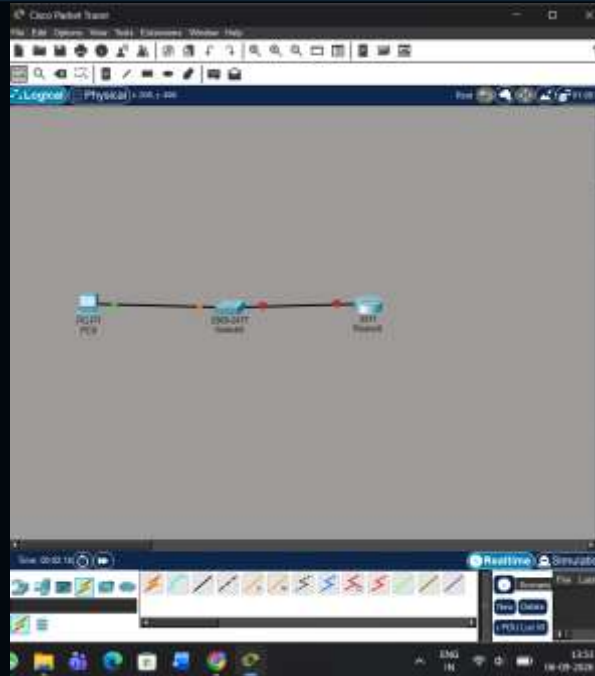


FIGURE 06 — Actual Packet Tracer topology evidence: PC0 → 2960-24TT Switch0 → 2911 Router0.

ADDRESSING PLAN

PC0

192.168.1.10

255.255.255.0

GATEWAY

192.168.1.1

Router Gi0/0 • 192.168.1.1 /24

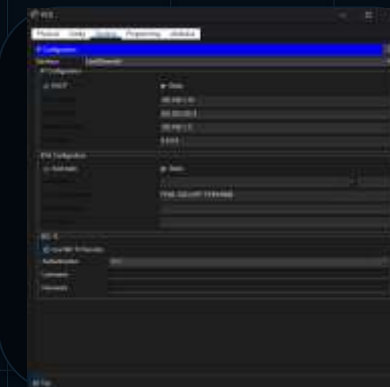


FIGURE 08 — PC0 static IPv4 configuration.

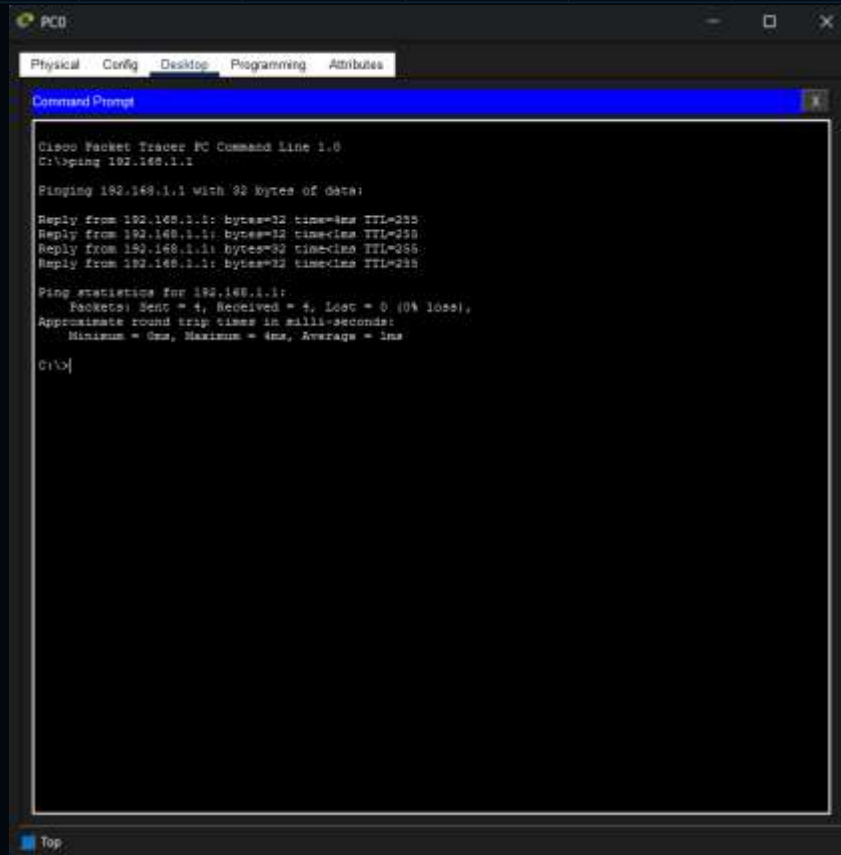


FIGURE 09 — Router CLI: Gi0/0 configured and verified up/up.

09 / VERIFICATION

Connectivity verification & learnings

The final practical check linked addressing, interface state, topology and reachability.



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=0ms TTL=255
Reply from 192.168.1.1: bytes=32 time=0ms TTL=255
Reply from 192.168.1.1: bytes=32 time=0ms TTL=255
Reply from 192.168.1.1: bytes=32 time=0ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Loss = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

SENT

4

RECEIVED

4

LOSS

0%

MIN

0 ms

MAX

4 ms

AVG

1 ms

KEY LEARNINGS

- Networking models provide a structured view of communication.
- Protocols serve different communication purposes.
- IPv4 addressing and subnetting help organize networks.
- Diagnostic commands provide practical visibility into connectivity and configuration.
- Packet Tracer connected theoretical addressing with an actual simulated topology.

FIGURE 10 — Actual Packet Tracer ping evidence: four replies from 192.168.1.1, 4 received, 0% loss; minimum 0 ms, maximum 4 ms, average 1 ms.

WEEK 1 COMPLETED • DG INTERNS HUB

10 / CONCLUSION

Conclusion

Week 1 brought the theory, command-line diagnostics and Packet Tracer practical into one connected workflow.

WEEK 1 CONCLUSION

I completed the Week 1 Networking Fundamentals scope by studying the required models, protocols and addressing concepts, then applying them through command-line diagnostics and a Cisco Packet Tracer connectivity test.

01	UNDERSTAND	OSI • TCP/IP • protocols
02	ADDRESS	IPv4 • masks • CIDR
03	OBSERVE	ping • ipconfig • ifconfig • tracet • traceroute
04	VERIFY	PC → switch → router • successful ping

FINAL TAKEAWAY

Networking became more concrete when definitions were connected to actual output and a working simulated topology.

PRACTICAL PROOF

4 replies • 0% loss
Average 1 ms

The evidence reflects the work actually completed.

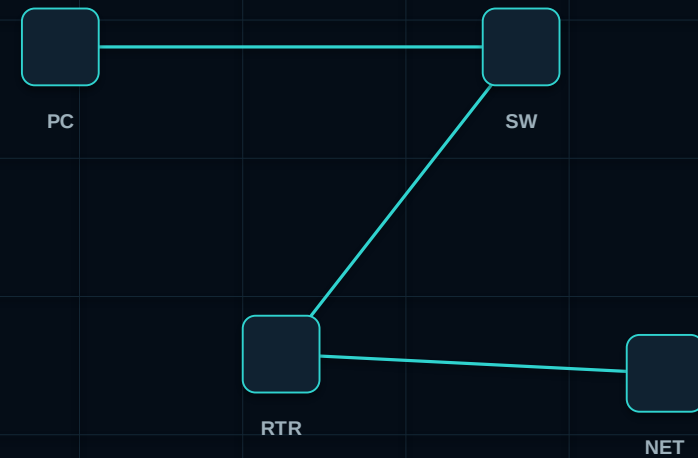
Thank You

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